

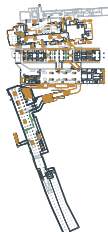
Arrival-time-aware evacuation analysis for a multi-line metro station

Longyang Road Station - dynamic network flow - continuous-space verification

1. INPUT

Station geometry and scenario definition

Station geometry



Longyang Road Station - DWG/DXF-derived layout

Demand and scenario inputs

Reference occupants + train arrivals
arrival-event timeline



Shared-resource parameters

Service μ_r

Receiving K_j

Speed-density $v(\rho)$

stairs / escalators / gates

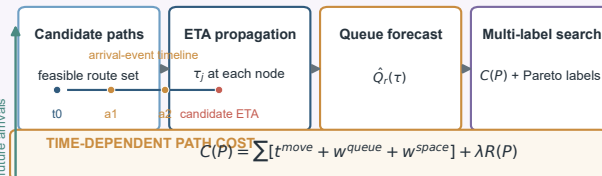
2. MODEL / OPTIMIZATION

Arrival-time-aware planning coupled to dynamic execution

CONTROLLED COMPARISON

Improved A*: current queue | Proposed AA*: queue at ETA | common executor

ARRIVAL-TIME-AWARE AA* ROUTING DECISION



Route assignment

rolling replanning

accepted requests

DYNAMIC NETWORK-FLOW EXECUTION

Shared-resource service
stairs / gates

Capacity + receiving
 μ_r, K_j

Density-speed update
 $v(\rho)$

In-transit events + route lock
state update

In-transit route locked - committed passengers keep their route

3. OUTPUT

Pathfinder verification and metrics

Export route & exit allocation

Pathfinder execution



continuous geometry - local interactions

Network layer

mean / T95 / T100
stationary exposure
load balance

Continuous space

completion time
congestion time
walking distance

Cross-model assessment

route transfer -> Pathfinder validation

